Hypocoercivity of Adaptive Langevin Dynamics

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Molecular Simulation homework by Jules Herrmann

Langevin Dynamics

$$\mathrm{d}q = M^{-1}p\,\mathrm{d}t$$
$$\mathrm{d}p = \left(-\nabla U(q) - \zeta M^{-1}p\right)\mathrm{d}t + \sigma\,\mathrm{d}W$$

- M : mass matrix
- ζ : friction
- σ : strengh of thermal noise

Invariant distribution = Boltzmann-Gibbs distribution $p_\beta \propto e^{-\beta U}$

With $\beta = \frac{2\zeta}{\sigma^2}$

Thermostat

∇U can be hard to compute.

We may only have access to a **noisy** approximation $\hat{\nabla} U$

Temperature of the system $T = \frac{1}{k_B \operatorname{Tr}(M^{-1})} \mathbb{E}(p^T M^{-2} p)$ must be constant.

Noise in $\hat{\nabla} U \rightarrow$ drift of system temperature $\rightarrow$ incorrect sampling

Solution : Control temperature with a thermostat [2]

Thermostat

Friction ζ reinterpreted as a dynamical variable

$$\mathrm{d}\zeta = \frac{1}{\nu}\left(p^T M^{-2} p - \frac{1}{\beta}\,\mathrm{Tr}(M^{-1})\right)$$

Temperature too low :

- $p^T M^{-2} p < \frac{1}{\beta}\,\mathrm{Tr}(M^{-1})$
- reduced friction
- less energy is dissipated
- temperature increases

Temperature too high :

- $p^T M^{-2} p > \frac{1}{\beta}\,\mathrm{Tr}(M^{-1})$
- increased friction
- more energy is dissipated
- temperature decreases

Adaptive Langevin Dynamics

$$
\begin{aligned}
\mathrm{d}q &= M^{-1} p \,\mathrm{d}t \\
\mathrm{d}p &= \left(-\nabla U(q) - \zeta M^{-1} p\right) \mathrm{d}t + \sigma_G \,\mathrm{d}W_G + \sigma_A \,\mathrm{d}W_A \\
\mathrm{d}\zeta &= \frac{1}{\nu}\left(p^T M^{-2} p - \frac{1}{\beta}\,\mathrm{Tr}(M^{-1})\right)
\end{aligned}
$$

- σ_G is the unknown noise of gradient approximation
- σ_A is the known thermal noise
- β is the prescribed inverse temperature
- ν is the strengh of the thermostat

Invariant distribution

$$\pi(\mathrm{d}q, \mathrm{d}p, \mathrm{d}\zeta) = Z^{-1} \exp\left(\frac{1}{\beta}\left[\frac{p^T M^{-1} p}{2} + U(q) + \frac{\nu}{2}(\zeta - \gamma)^2\right]\right) \mathrm{d}q\, \mathrm{d}p\, \mathrm{d}\zeta$$

Where

- Z is such that this distribution is a probability measure
- $\gamma = \frac{\beta(\sigma_A^2 + \sigma_G^2)}{2}$

Normalized Dynamics

Change of variable $\varepsilon = \sqrt{\nu}$ and $\xi = \sqrt{\nu}(\zeta - \gamma)$

$$\mathrm{d}q = p\,\mathrm{d}t$$

$$\mathrm{d}p = \left(-\nabla U(q) - \left(\frac{\xi}{\varepsilon} + \gamma\right)p\right)\mathrm{d}t + \sqrt{\frac{2\gamma}{\beta}}\,\mathrm{d}W$$

$$\mathrm{d}\xi = \frac{1}{\varepsilon}\left(p^T p - \frac{n}{\beta}\right)$$

Normalized Dynamics

New normalized invariant distribution

$$\pi(\mathrm{d}q, \mathrm{d}p, \mathrm{d}\xi) = Z^{-1} \exp\left(\frac{1}{\beta}\left[\frac{p^T p}{2} + U(q) + \frac{\xi^2}{2}\right]\right) \mathrm{d}q\, \mathrm{d}p\, \mathrm{d}\xi$$

Independant from γ and ε : we can more easily study the effect of the parameters of the system on its convergence

Trajectory average

Langevin dynamics are used to estimate $\mathbb{E}_\pi(\varphi)$ with the trajectory average estimator $\hat{\varphi}_t$

$$\hat{\varphi}_t = \frac{1}{t}\int_0^t \varphi(q_s, p_s, \xi_s)\,\mathrm{d}s$$

π is invariant, the system is irreducible and regular :

$$\hat{\varphi}_t \xrightarrow[t\to\inf]{\text{a.s.}} \mathbb{E}_\pi(\varphi)$$

Central Limit Theorem

$$\sqrt{t}(\hat{\varphi}_t - \mathbb{E}_\pi(\varphi)) \xrightarrow[t\to\infty]{\text{law}} \mathcal{N}\big(0, \sigma^2_{\varepsilon,\gamma}(\varphi)\big)$$

And there exist C and λ, s.t.

$$\sigma^2_{\varepsilon,\gamma}(\varphi) \leq \frac{2C\ \|\varphi\|^2_{L^2(\pi)}}{\lambda}\ \max\left(\gamma, \frac{\varepsilon^2}{\gamma}, \gamma\varepsilon^2, \frac{1}{\gamma\varepsilon^2}\right)$$

Numerical Integrator

Generator	Single Numerical Step
$\mathcal{L}_A = p \cdot \nabla_q$	$\Phi^A_{\Delta_t}(q) = q + \Delta t p$
$\mathcal{L}_B = -\nabla U(q) \cdot \nabla_p$	$\Phi^B_{\Delta_t}(p) = p - \Delta t \nabla U(q)$
$\mathcal{L}_D = \frac{1}{\varepsilon}\left(\lvert p\rvert^2 - \frac{n}{\beta}\right)\partial_\xi$	$\Phi^D_{\Delta_t}(\xi) = \xi + \frac{\Delta t}{\varepsilon}\left(\lvert p\rvert^2 - \frac{n}{\beta}\right)$
$\mathcal{L}_O = -\zeta p \cdot \nabla_p + \sigma^2 \Delta_p$	$\Phi^O_{\Delta_t}(p) = e^{-\Delta t \zeta} p + G(\sigma, \zeta, \Delta t) R$

Where $R \sim \mathcal{N}(0,1)$ and $G(\sigma, \zeta, \Delta t) = \begin{cases} \sigma\sqrt{\frac{1-e^{-2\Delta t \zeta}}{2\zeta}} & \text{if } \zeta \neq 0 \\ \sigma\sqrt{\Delta t} & \text{if } \zeta = 0 \end{cases}$

Numerical Integrator

- BADODAB

$$\Phi^B_{\frac{\Delta t}{2}} \circ \Phi^A_{\frac{\Delta t}{2}} \circ \Phi^D_{\frac{\Delta t}{2}} \circ \Phi^O_{\Delta t} \circ \Phi^D_{\frac{\Delta t}{2}} \circ \Phi^A_{\frac{\Delta t}{2}} \circ \Phi^B_{\frac{\Delta t}{2}}$$

- ODABADO

$$\Phi^O_{\frac{\Delta t}{2}} \circ \Phi^D_{\frac{\Delta t}{2}} \circ \Phi^A_{\frac{\Delta t}{2}} \circ \Phi^B_{\Delta t} \circ \Phi^A_{\frac{\Delta t}{2}} \circ \Phi^D_{\frac{\Delta t}{2}} \circ \Phi^O_{\frac{\Delta t}{2}}$$

CLT on toy example

$U(q) = \left(q^2 - 1\right)^2 + \frac{1}{2}q$

$$\hat{\varphi}_K = \frac{1}{K}\sum_{k=0}^{K-1} \varphi\big(q^{(k)}, p^{(k)}, \xi^{(k)}\big)$$

$$\bar{\varphi}_{K,N} = \frac{1}{N}\sum_{n=1}^{N} \hat{\varphi}_K^{(n)}$$

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{n=1}^{N}\Big(\hat{\varphi}_K^{(n)} - \bar{\varphi}_{K,N}\Big)$$

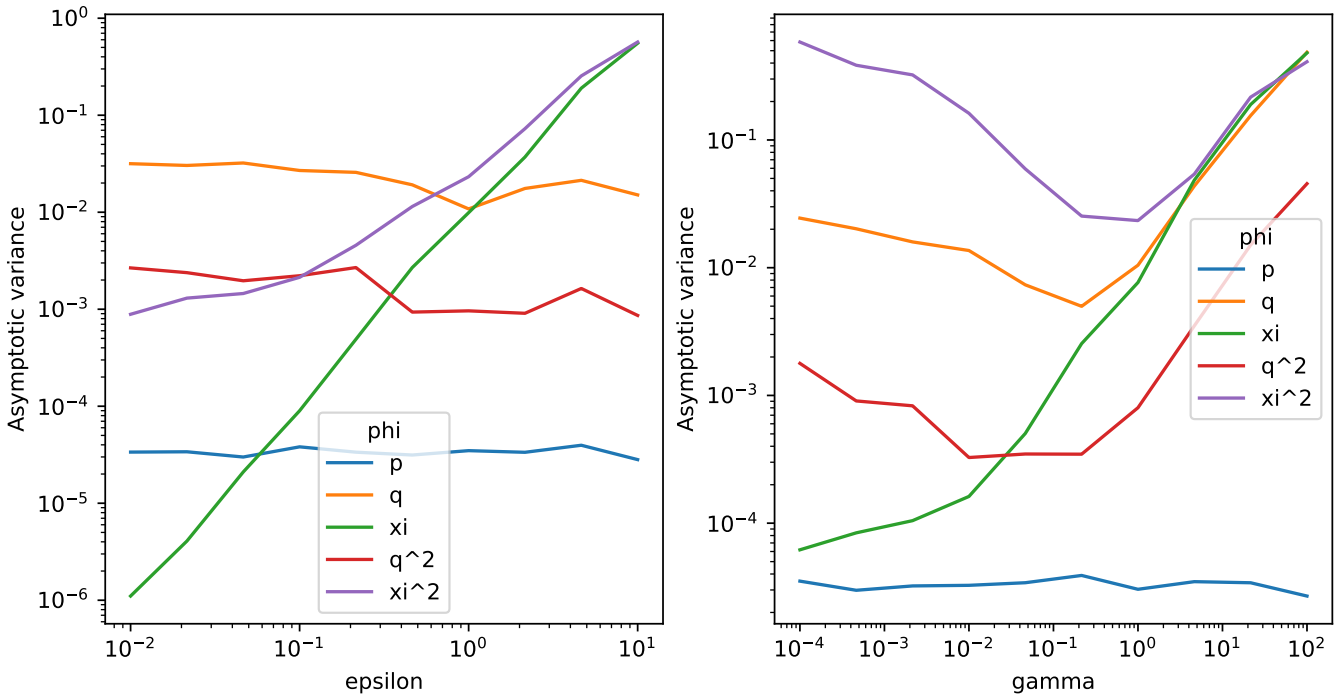

Figure 1: Variance of different observable (Right : $\gamma = 1$, Left : $\varepsilon = 1$)

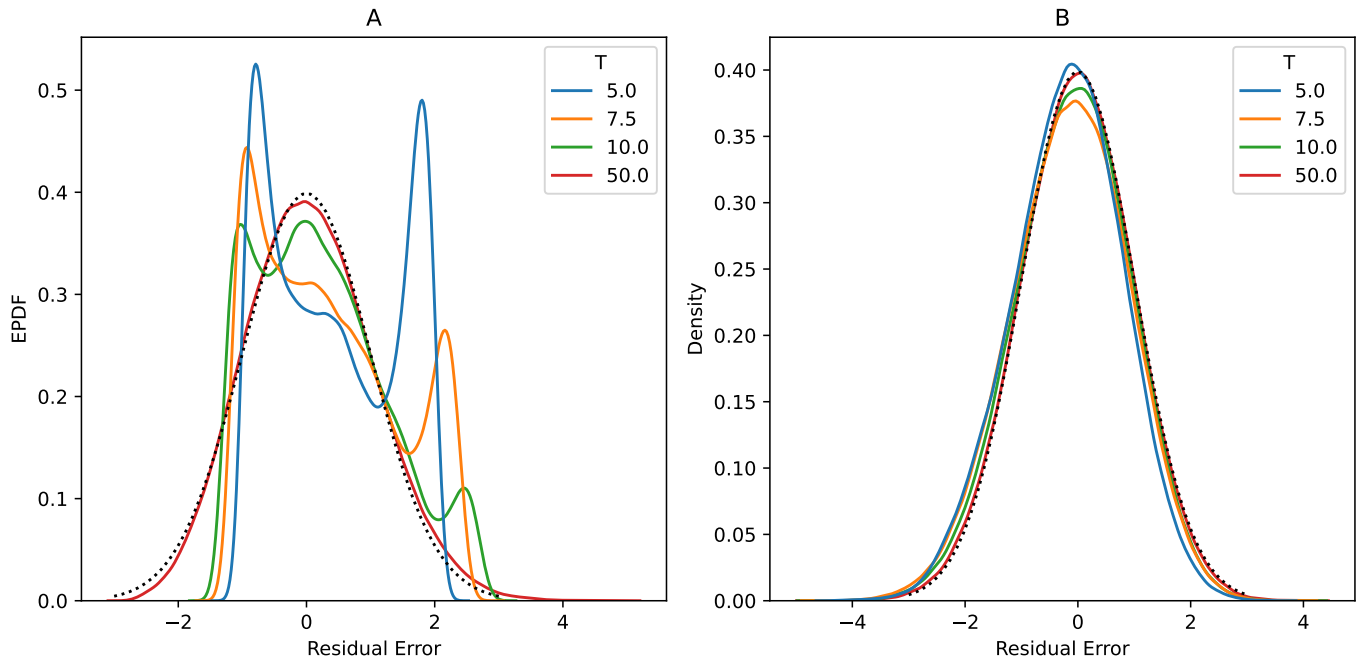

Figure 2: Empirical distribution of rescaled residual error at different times

Application to Bayesian Logistic Regression with a Stochastic gradient

Observation : $y_j \in \{0,1\}$, $x_j \in \mathbb{R}^d$ for $j \in \left[0, \tilde{N}\right]$

Parameter : $q \in \mathbb{R}^d$

Prior : $q \sim \mathcal{N}\left(0, \sigma^2\right)$ i.e. $p_0(q) \propto \exp\left(-\frac{|q|^2}{\sigma}\right)$

Likelihood : $p\left(\left(y_j\right), \left(x_j\right) \mid q\right) = \prod_j \frac{\exp\left(y_j < x_j, q >\right)}{1 + \exp\left(< x_j, q >\right)}$

Posterior : $p\left(q \mid \left(y_j\right), \left(x_j\right)\right) \propto p_0(q)\ p\left(\left(y_j\right), \left(x_j\right) \mid q\right)$

If we choose $-\nabla U(q) = \nabla \log(p_0(q)) + \nabla \log\big(p\big((y_j), (x_j) \mid q\big)\big)$

Then, the invariant distribution is the posterior distribution

$$\pi(\mathrm{d}q) \propto \exp(-U(q))\,\mathrm{d}q \propto p\big(q \mid y_j, x_j\big)\,\mathrm{d}q$$

$-\nabla U(q)$ expensive to compute : use of a stochastic gradient

$$-\tilde{\nabla} U = \nabla \log(p_0(q)) + \frac{\tilde{N}}{m} \sum_{j \in B} \nabla \log\big(p\big(y_j, x_j \mid q\big)\big)$$

With B a random batch of size m of observations

Application to MNIST dataset, projected on a smaller space using a Principal Component Analysis

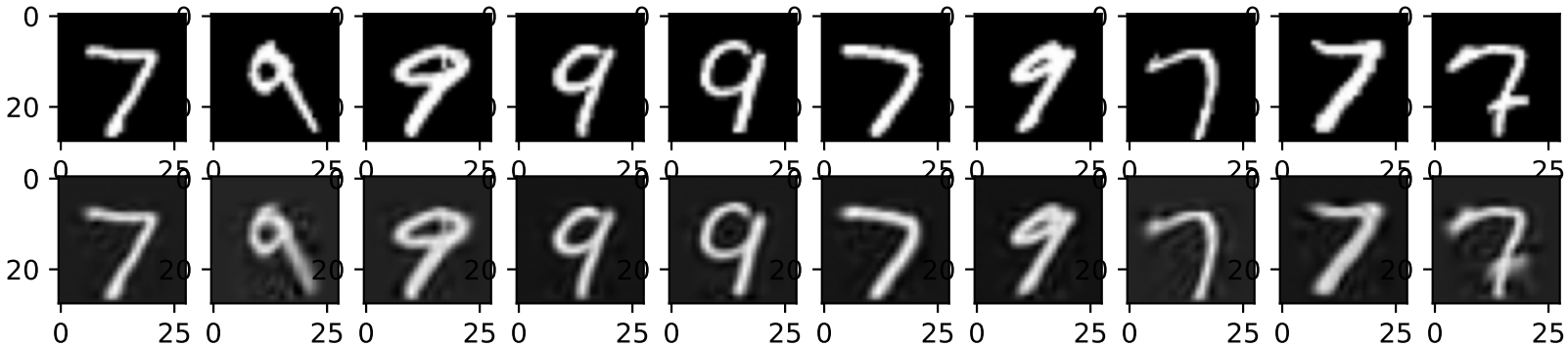

Figure 3: Examples of images from the MNIST dataset and their projection on the 100 principal components

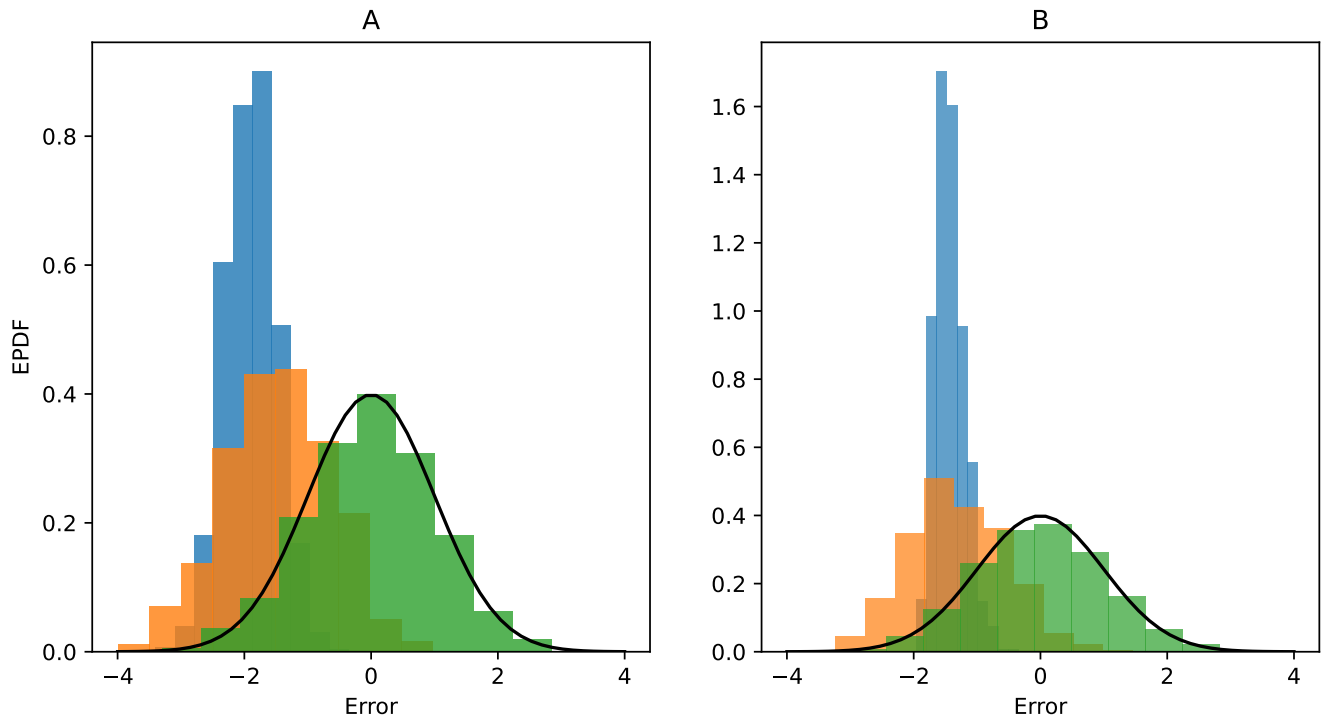

Figure 4: Empirical distribution of error (Right : q_{65}, Left : q_{65}^2)

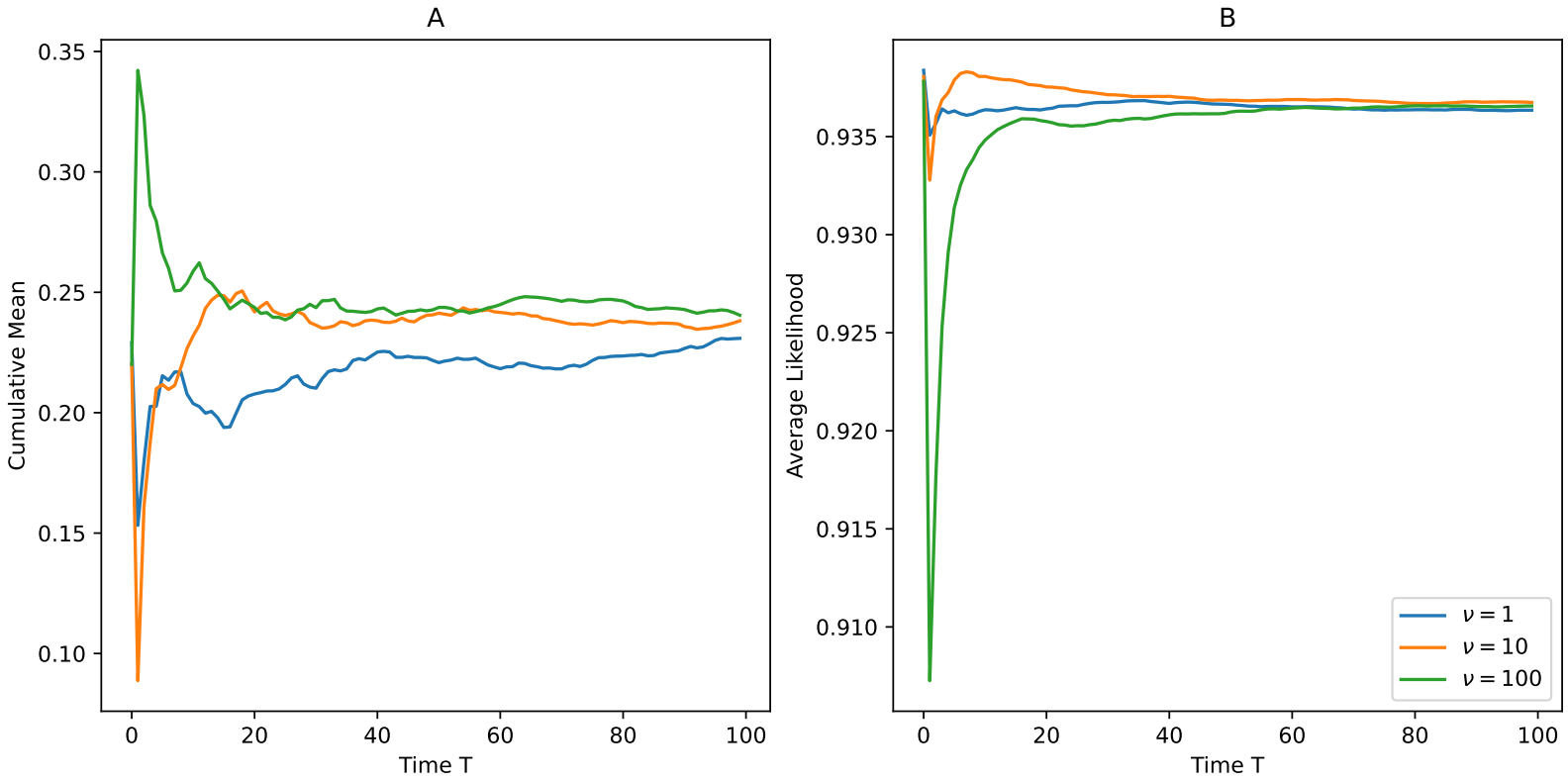

Figure 5: Evolution of $\mathbb{E}(q_{65})$ and of the average likelihood over time

[1] B. Leimkuhler, M. Sachs, and G. Stoltz, “Hypocoercivity properties of adaptive Langevin dynamics.” [Online]. Available: https://arxiv.org/abs/1908.09363

[2] N. Ding, Y. Fang, R. Babbush, C. Chen, R. D. Skeel, and H. Neven, “Bayesian sampling using stochastic gradient thermostats,” *Advances in neural information processing systems*, vol. 27, 2014.